

There are two parameters which are num1 and num2, and their data type is long. The initial values of them are 2 and -5. The ciphertext is the encrypted message.

main

movl num1(%rip), %eax means load the num1 in the memory to the %eax, and then %eax is 2
movl num2(%rip), %ecx means load the num2 in the memory to the %ecx, and then %ecx is -5
addl %ecx, %eax means $\%eax = \%eax + \%ecx$
movb %al, %bl select the lower 8 bit of %eax and put them into %bl
So the key is 0xFD, in decimal value is -3

decrypt_loop

movzbq (%rdi), %rax means read the current address and store it to the %rax
testb %al, %al means to check the string is 0 or not
je decrypt_done means if previous step tests, jump to decrypt_done.
subb %bl, %al means $\%al = \%al - \%bl$
movb %al, (%rdi) means move %al which has been solved to the %rdi.
inc %rdi means %rdi increase 1, solve the next string.
jmp decrypt_loop means jump to the head of this loop.

We need to transfer the text to ASCII, the function of whole can be *ASCII of result = ASCII of secret +3*

secret	ASCII	+3	result
d	100	103	g
b	98	101	e
w	119	122	z

RESULT: “ggez”